A **polynomial** is an algebraic expression that consists of terms where variables are raised to non-negative integer powers and combined using addition, subtraction, and multiplication. Importantly, it does **not** involve division by any variables.

For example:

$$P(x) = 3x^2 + 2x + 1$$

is a polynomial, since it only involves terms with variables raised to non-negative integer powers and no division by variables.

A **rational expression** is an algebraic expression that can involve division by a polynomial. The **numerator** and **denominator** of a rational expression may both be polynomials, but the overall expression involves a division by a variable.

For example:

$$R(x) = \frac{3x^2 + 2x + 1}{x + 1}$$

is a rational expression because it involves division by $x + 1$, which is a polynomial. However, this expression is **not** a polynomial due to the division by a variable in the denominator.